

2(a)(i)

A = adenine
T = thymine **R** thiamine / thyamine
G = guanine
C = cytosine

all correct = 2 marks
one, two or three correct = 1 mark

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Question	Answer	Marks
2(a)(ii)	<i>three from:</i> 1 complementary base pairing / complementary bases / base pairing / base pairs / complementary pairing ; 2 A with T <u>and</u> G with C ; 3 hydrogen bonding / hydrogen bonds ; <i>in context of between base pairs / holding strands together</i> 4 ref. to purines with pyrimidines ; 5 double ring (bases) with single ring (bases) ; A <i>idea of</i> longer base with shorter base 6 two (DNA), strands / polynucleotides ; A two chains A double helix (as double implies two strands) 7 strands (anti)parallel / distance between strands always the same ;	3
2(b)	deoxyribose ; A 2-deoxyribose / 2-deoxy-D-ribose	1
2(c)	<i>two from:</i> (information could be) sequence of <u>amino acids</u> ; A idea that a polypeptide/protein has amino acids arranged in an order / AW I primary structure / chains of amino acids (different) proteins have, different / specific, sequences (of amino acids) ; (up to) 20 different amino acids in proteins ; A approximately 20 <i>idea that</i> there is a great variety in protein structure ; I have many functions	2